MIL-A-25367B (USAF)
25 MAY 1988
SUPERSEDES
MIL-A-25367A(USAF)
2 JANUARY 1958

MILITARY SPECIFICATION

ANTENNA AT-536/ARN

This specification is approved for use within the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 This specification establishes the requirements for a cavity type, flush-mounted, VHF marker beacon receiving antenna designated Antenna AT-536/ARN.

2. APPLICABLE DOCUMENTS:

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Military

MIL-E-5400	Electronic Equipment, Aerospace, General Specification For
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of
MIL-C-39012	Connector, Coaxial, Radio Frequency, General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: WR-ALC/MMIRG, Robins AFB, GA 31098-5609, by using the self-addressed standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5826

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STANDARDS

Military

MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-831	Test Reports, Preparation of

2.1.2 Other Government documents. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. Air Force

55D13159	Antenna AT-536/ARN Outline Dimensions of

(Copies of specifications, standards, and other Governments documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail) (specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 General Specification. The requirements of MIL-E-5400 for class 2 equipment apply as requirements of this specification with the exceptions or additions called out herein. When the two specifications conflict, this specification shall govern.

3.2.1 Connectors. The antenna connector shall be designed to accommodate Plug UG 88/U in accordance with MIL-C-39012.

3.2.2 Temperature. The antenna shall withstand a temperature of 250 degrees F. without electrical or mechanical deterioration.

3.3 Design.

3.3.1 General. The general design of the antenna shall be as shown on Drawing 55D13159.

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3.3.2 Dimensions. Envelope dimensions and mounting dimensions shall be as specified on Drawing 55D13159.

3.4 Sealing. The utmost care shall be used in manufacturing the housing and surface plate. These shall be cemented together, and all openings in the cavity wall cemented so as to provide an adequate pressure seal. An adequate seal is considered to be one in which no continuous flow of bubbles occurs while the antenna is subjected to an atmospheric pressure (while submerged in water) simulating an altitude of 70,000 feet, for at least 2 minutes.

3.5 Electrical Requirements.

3.5.1 Frequency. The antenna shall be tuned to the exact frequency of 75 mc by means of a crystal-controlled signal source and then sealed so as to prevent further tuning adjustment.

3.5.2 Impedance. The input impedance of the antenna at its input socket shall be such as to cause a voltage standing wave ratio (VSWR) of 1.5 to 1 or less at 75.0 mc on its connecting cable (RG-58/U) with the antenna mounted at the center of a flat ground plane (or curved to match the antenna surface) at least 8 feet by 8 feet. A bandwidth of 300 kc shall be maintained within a 5 to 1 VSWR circle with the antenna fixed tuned.

3.5.3 Radiation Pattern. The antenna shall be essentially horizontally polarized and oriented to be parallel with the largest dimensions of the antenna cavity. The direction of maximum radiation shall be within 10 degrees of the perpendicular to the face of the antenna, and the included angle between the half-power points of the maximum lobe shall be not less than 100 degrees.

3.5.4 Efficiency. The minimum acceptable sensitivity (in voltage) of the test antenna shall be within 10.5 db of that of a half-wave dipole at a frequency of 75 megacycles.

3.6 Weight. Weight of the antenna shall be minimum in keeping with the other requirements of this specification and shall not exceed 2 pounds and 12 ounces.

3.7 Mounting. Mounting holes shall be drilled into the flange of the antenna according to Drawing 55D13159.

3.8 Identification of Product. Identification marking of the antenna selector shall be in accordance with MIL-STD-130 and shall contain the following information.

Nomenclature
Stock number
Serial number
Manufacturer's name or code

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4. QUALITY ASSURANCE PROVISIONS:

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of Tests. The inspection and testing of the antenna shall be classified as follows:

a. Preproduction Tests . . . . . See 4.4

b. Acceptance Tests . . . . . See 4.5

4.3 Test Conditions. Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of approximately 29.92 inches of mercury and a temperature of approximately 25°C (77°F).

4.4 Preproduction Tests.

4.4.1 Preproduction Test Samples. When specified in the contract, a minimum of three antennas shall be subjected to the preproduction and the acceptance tests specified herein:

4.4.1.1 Samples For Procuring Activity. Three antennas shall be submitted to the procuring activity at the time the manufacturer submits the preproduction test report for approval. These antennas shall be used by the Government for:

a. A review of the electrical characteristics and mechanical construction of the antenna.

b. The performance of any tests deemed necessary after a review of the manufacturer's report of tests on the preproduction sample.

c. Government reference purposes.

4.4.2 Preproduction Test Program. Prior to any formal preproduction testing of the sample antennas, the manufacturer shall submit a draft of his proposed test program for approval to the procuring activity. The test program shall include the procedures and equipment used in performing the preproduction tests specified herein.

4.4.3 Preproduction Test Report. After completion of the preproduction tests by the manufacturer, three copies of the complete report shall be furnished the procuring activity. The test report shall be in accordance with MIL-STD-831 and shall include the name and grade of all materials used in the antenna.

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4.4.4 Tests.

4.4.4.1 Environmental Tests. The sample antennas shall be subjected to environmental tests in accordance with the following procedures specified in MIL-STD-810.

a. Humidity, Method 507, Procedure I.

b. Vibration, Method 514.2-II, Procedure I, Curve E (15g) (5-2000Hz).

c. Shock, Method 516, Procedure I, Half size wave, 15g's peak, 11Ms duration, operation not required during test.

d. Salt spray, Method 509, Procedure I.

e. Temperature shock, Method 503, Procedure I.

4.4.4.1.1 Environmental Effects. During all environmental testing, the VSWR of the antenna shall not go beyond 2.5 when measured under the conditions outlined in 3.5.2.

4.4.4.1.2 High Temperature Tests. The sample antenna shall be placed within a chamber and the internal temperature shall be raised to 250 degrees F. with an internal relative humidity of not more than 5 percent. The antenna shall remain in this environment for 4 hours, at the end of which time the antenna shall be removed from the chamber and immediately measured for VSWR.

4.4.4.2 Radiation Pattern. The test setup of figure 1 shall be used with a spacing "D" of 30 feet between the transmitting and receiving antennas. Radiation pattern data shall be obtained by recording the field strength meter output at intervals of 15 degrees as the ground plane containing the test antenna is rotated 360 degrees around a vertical axis through the center of the test antenna. Pattern data shall be taken for two conditions: one with the ground plane so mounted that the antenna radiating element is vertical, and the other with the ground plane so mounted that the antenna element is horizontal.

4.4.4.3 Efficiency. The test setup of figure 1 shall be used with a spacing "D" of 30 feet between the transmitting and receiving antennas and employing vertical polarization. A receiving effectiveness comparison shall be made between the test antenna mounted in the ground plane, and a 74-inch dipole antenna placed in the same physical location but with the ground plane removed, by observing the relative signal attenuator settings for a fixed output reading of the field strength meter or receiver.

4.5 Acceptance Tests. Acceptance tests shall consist of individual tests and sampling tests.

4.5.1 Individual Tests. Each antenna shall be subjected to any other test considered necessary to determine compliance with the requirements of this specification.

4.5.1.1 Visual Inspection. Each antenna shall be inspected for uniform quality, injurious defects in the material, and proper finish.

4.5.1.2 Impedance. The VSWR of each antenna shall be measured under the conditions stated in 3.5.2. Measurement shall be made by use of an approved slotted line or equivalent. Prior approval of the impedance measuring equipment used is required.

4.5.2 Sampling Tests. Two antennas shall be selected at random from each quantity of 100 or fraction thereof on the order and subjected to the sampling tests. These tests shall be in addition to the individual tests specified herein.

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4.5.2.1 Mechanical Inspection. The antenna shall be subjected to such mechanical inspection and tests as the procuring activity deems necessary or desirable to determine compliance with the requirements of this specification. This includes checking of the outside diameter, mounting holes, cable connection and general workmanship of the antenna.

4.5.2.2 Additional Impedance Tests. A check or bandwidth test shall be made. In addition, the antenna may be subjected to such electrical tests as the procuring activity deems necessary or desirable to determine compliance with the requirements of this specification.

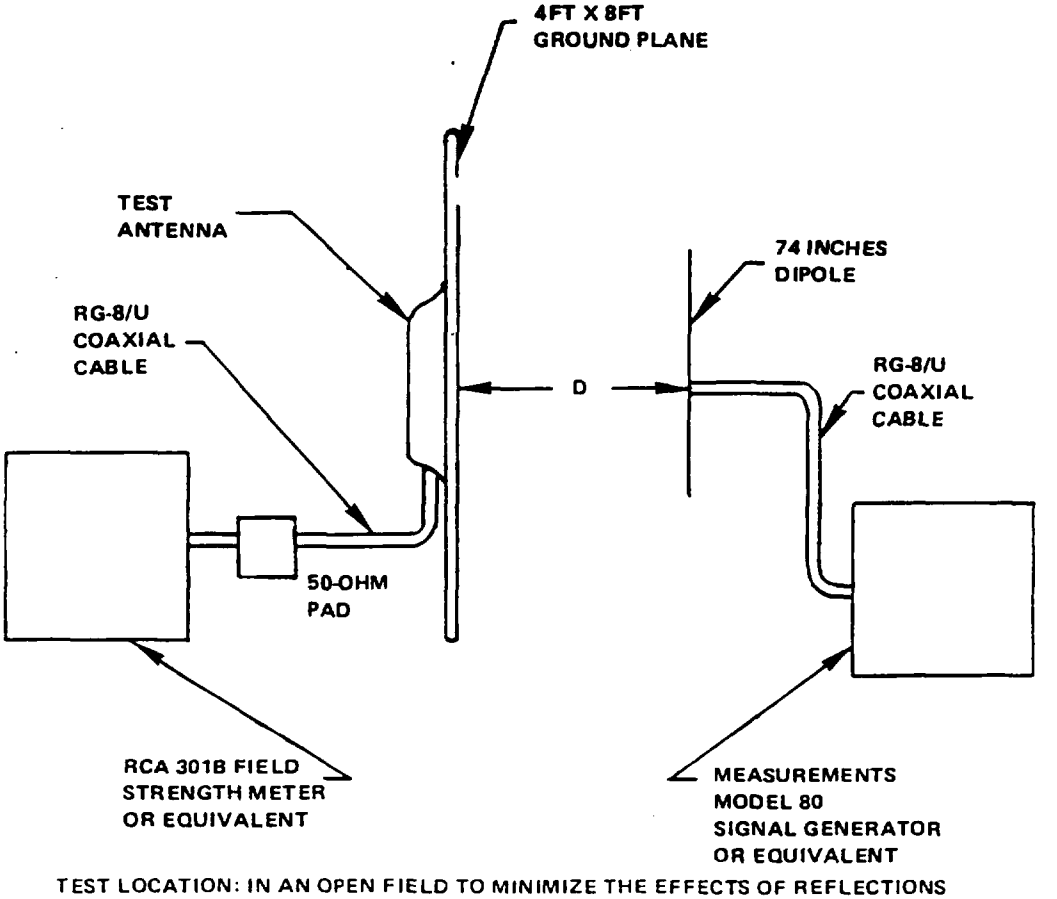
4.5.2.3 Seal Test. Leakage (Immersion), Method 512, Procedure III. The antenna shall be subjected to an immersion test performed IAW MIL-STD-810. A vacuum pressure of 1.3 inches of hg. shall be maintained for 1 minute in performance of test (Para 3.3.2). The pressure shall be increased to 2.5 inches of hg and maintained undisturbed for 10 minutes.

4.5.2.4 Acceptance Criteria. When one or more items fail to meet the specification, acceptance of all items will be withheld until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

5. PREPARATION FOR DELIVERY:

5.1 General. The antenna shall be preserved, packaged, packed, and marked for shipment in accordance with MIL-E-17555.

5.2 Shipment Marking. The shipment marking nomenclature shall be: Antenna AT-536/ARN.



TEST LOCATION: IN AN OPEN FIELD TO MINIMIZE THE EFFECTS OF REFLECTIONS

FIGURE 1. SENSITIVITY AND SELECTIVITY TEST SETUP

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6. NOTES:

6.1 Intended Use. The antenna is intended for airborne use as a receiving antenna for Marker Beacon Receiving Set AN/ARN-32.

6.2 Ordering Data. Procurement documents should state the levels of preservation, packaging and packing required, and the conditions for the following:

6.2.1 Preproduction Tests. It is expected that the contract or purchase order will specify that a minimum of three antennas will be required as preproduction samples to be subjected to the preproduction tests to determine compliance with the requirements of this specification and that three antennas will be required as preproduction samples to be submitted to the procuring activity. The procurement document should specify the point of inspection for these tests.

6.3 Subject Term (Key Word) Listing:

Antenna
ARN
AT-536

Custodian:

Air Force - 99

Preparing Activity:

Air Force - 84

Project No:
5826-F193

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

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